

Quiz 2

Name:

Student ID:

Set: A B C D (circle one)

- Time allowed: 20 min

1. (2.5 marks) Use rules of inference to prove that the following argument is valid.

Note: You must use the provided rules of inference and list the rule. No marks will be given for proofs that involve a truth table (i.e. critical rows).

	STEPS	REASON
$\neg w \vee \neg d$	1) $w \wedge x \wedge h$	given
$w \wedge x \wedge h$	2) w	C.S. 1
$(a \vee \neg z) \rightarrow d$	3) $\neg w \vee \neg d$	given
$\therefore z$	4) $\neg d$	D.S. 2,3
	5) $(a \vee \neg z) \rightarrow d$	given
	6) $\neg(a \vee \neg z)$	M.T. 4,5
	7) $\neg a \wedge \neg \neg z$	De Morgan's 6
	8) $\neg a \wedge z$	Double negation 7
	9) z	C.S. 8
	10)	
	11)	
	12)	
	13)	
	14)	

2. (2.5 marks)

Circle either TRUE or FALSE and justify your answer with a table of values.

No marks will be given for answers without justification.

Marks will be deducted for unnecessary rows in the table.

Consider the following sets:

$$A = \{-1, 1, 2, 6, 7, 10, 14\}$$

$$B = \{0, 3, 4\}$$

a. $\exists x \in A, \forall y \in B$ such that $x - y \in A$

TRUE / FALSE

x	y	$x - y \in A$
10	0	$10 \in A \checkmark$
10	3	$7 \in A \checkmark$
10	4	$6 \in A \checkmark$

b. $\forall x \in A, \exists y \in B$ such that $x + y \notin A$

TRUE / FALSE

x	y	$x + y \notin A$
-1	4	$3 \notin A \checkmark$
1	3	$4 \notin A \checkmark$
2	3	$5 \notin A \checkmark$
6	3	$9 \notin A \checkmark$
7	4	$11 \notin A \checkmark$
10	3	$13 \notin A \checkmark$
14	3	$17 \notin A \checkmark$